AWURAMA OFORI-KWAFO

PhD Candidate, Biomedical Engineering, University of Toledo

Curriculum Vitae | Translational research | Microphysiological systems | Cellular interactions

RESEARCH PROFILE

I am a doctoral candidate in Biomedical Engineering at the University of Toledo. My research applies microphysiological systems to the study of cellular interactions, with a translational focus. I integrate microfluidic engineering, primary cell culture, and quantitative functional assays to examine how cell populations communicate under controlled, physiologically relevant conditions.

My work has resulted in four peer-reviewed publications, two as first author, and presentations at national scientific meetings. I am also committed to teaching, mentorship, and STEM outreach.

EDUCATION

Doctor of Philosophy (PhD), Biomedical Engineering *January 2021 to Present*
University of Toledo, Toledo, Ohio · Advisor: Dr. Yuan Tang

M.S. in Bioengineering *August 2026*
University of Toledo, Toledo, Ohio

Bachelor of Science (B.Sc.), Biomedical Engineering *2017*
Kwame Nkrumah University of Science and Technology, Kumasi, Ghana

PUBLICATIONS

Peer-reviewed journal articles

1. **Butyrate triggers neuronal signaling via release of colonic serotonin in a microfluidic model of gut-brain axis**
   Ofori-Kwafo, A., Sree Kumar, H., Otmanowski, E.B., Al Mamun, E., Habibic-Beskovic, A., Li, Q., Zubcevic, J., & Tang, Y. (2026)
   American Journal of Physiology-Gastrointestinal and Liver Physiology
   First author · online ahead of print
   DOI: https://journals.physiology.org/doi/abs/10.1152/ajpgi.00014.2026
2. **Gut-on-a-chip platforms: Bridging in vitro and in vivo models for advanced gastrointestinal research**
   Ofori-Kwafo, A., Sigdel, I., Al Mamun, E., Zubcevic, J., & Tang, Y. (2025)
   Physiological Reports, 13, e70356
   First author · comprehensive review of microfluidic gut-on-a-chip technology
   DOI: https://doi.org/10.14814/phy2.70356
3. **Protein Kinase C-Delta (PKCδ) inhibition stabilizes endothelium and suppresses triple-negative breast cancer (TNBC) intravasation in a microfluidic hypoxic tumor model**
   Sigdel, I., Ofori-Kwafo, A., Al Mamun, E., Tiwari, A.K., & Tang, Y. (2025)
   Bioengineering & Translational Medicine, e70090
   DOI: https://doi.org/10.1002/btm2.70090
4. **Biomimetic on-chip assay reveals the anti-metastatic potential of a novel thienopyrimidine compound in triple-negative breast cancer cell lines**
   Sigdel, I., Ofori-Kwafo, A., Heizelman, R.J. III, Nestor-Kalinoski, A., Prabhakarpandian, B., Tiwari, A.K., & Tang, Y. (2023)
   Frontiers in Bioengineering and Biotechnology, 11, 1227119
   DOI: https://doi.org/10.3389/fbioe.2023.1227119

CONFERENCE PRESENTATIONS

Oral presentations

2026 Ofori-Kwafo, A., Sree Kumar, H., Al Mamun, E., Zubcevic, J., & Tang, Y. A Gut-on-a-Chip Model of the Gut-Brain Axis Reveals Butyrate-Driven Serotonergic Epithelial-Neuronal Signaling.
Biomedical Engineering Society (BMES) Annual Meeting, Orlando, Florida, October 21 to 24, 2026 (upcoming)

2023 Ofori-Kwafo, A., Sigdel, I., Khalaf, K., Zubcevic, J., & Tang, Y. Simulating the Effect of the Gut Microbiome on the Gut-Brain Axis on a Chip.
Biomedical Engineering Society (BMES) Annual Meeting, Seattle, Washington, October 11 to 14, 2023

Poster presentations

2026 Ofori-Kwafo, A., Sree Kumar, H., Al Mamun, E., Habibic-Beskovic, A., Li, Q., Zubcevic, J., & Tang, Y. Microbial Metabolite Butyrate Activates Sensory Neurons through Colonic Serotonin Release in a Microfluidic Co-Culture Model.
American Physiological Society (APS) Annual Meeting, Minneapolis, Minnesota, April 23 to 26, 2026

2024 Ofori-Kwafo, A., Zubcevic, J., & Tang, Y. Establishing a Tri-Culture Model to Study the Interactions of the Gut-Brain Axis on a Microfluidic Device.
BME Research Day, University of Akron, Akron, Ohio, March 15, 2024

2022 Ofori-Kwafo, A., Sigdel, I., Heizelman, R.J. III, & Tang, Y. Quantifying Breast Cancer Intravasation, Extravasation and Angiogenesis Using High Throughput Assays.
Biomedical Engineering Society (BMES) Annual Meeting, San Antonio, Texas, October 12 to 15, 2022

2021 Tang, Y., Sigdel, I., Faizee, F., Ofori-Kwafo, A., & Tiwari, A.K. Preventing Breast Cancer Metastasis by Attenuating Chemotherapy-Induced Endothelial Inflammation.
Biomedical Engineering Society (BMES) Annual Meeting, Orlando, Florida, October 6 to 9, 2021

RESEARCH EXPERIENCE

Graduate Research Assistant, Department of Bioengineering *January 2021 to Present*
University of Toledo, Biofluidics Lab, Toledo, Ohio

- Engineered a tri-culture microfluidic model of the gut-brain axis that integrates gut microbes, epithelial cells, and sensory neurons, enabling real-time study of host-microbiome signaling. Operated and maintained more than 200 microfluidic devices.
- Independently established a high-throughput microfluidic model of breast cancer metastasis (first-author poster, BMES 2022) that combines endothelial monolayers, collagen and Matrigel gels, and a drug-resistant triple-negative breast cancer line to model intravasation, extravasation, and angiogenesis, and characterized endothelial barrier breakdown, invasion, and sprouting by VE-cadherin staining and imaging.
- Maintained four or more primary and established cell types, including colonic epithelial cells, sensory neurons, endothelial cells, and cancer cell lines, and adapted protocols across standard and microfluidic platforms.
- Designed co-culture experiments that reproduce physiological tissue microenvironments and generate reproducible data for mechanistic studies.
- Quantified epithelial barrier integrity and neuronal excitability using permeability assays, tight junction characterization, and real-time calcium imaging with single-cell analysis.
- Characterized protein and gene expression across multi-cell-type models using immunofluorescence, Western blot, PCR and qPCR, and RNA extraction.
- Applied quantitative image analysis at single-cell resolution to assess cell morphology, barrier integrity, and neuronal activity, contributing data to four peer-reviewed publications.
- Developed and validated a transfection protocol for primary colonic epithelial cells within a microfluidic device, confirming expression by fluorescence imaging.

- Performed cytokine profiling and ELISA to quantify inflammatory mediator secretion in co-culture systems under pharmacological stimulation.
- Conducted reactive oxygen species, cell viability, and dose-response assays to characterize compound activity in microfluidic models.
- Maintained long-term perfusion cultures under defined flow and dosing conditions, ensuring reproducibility across multi-week studies.
- Developed a blood-brain barrier model incorporating brain microvascular endothelial cells and pericytes as a secondary research platform for the laboratory.
- Mentored undergraduate and graduate students in device operation, cell culture, imaging, and experimental design.

TEACHING EXPERIENCE

Graduate Teaching Assistant, College of Engineering *January 2021 to Present*

University of Toledo, Department of Bioengineering · 4 to 5 semesters across multiple courses, including two laboratory courses · 100+ undergraduate students instructed

BIOE 3500: Bioprocessing Laboratory

- Instructed 20 to 30 undergraduates per semester in recombinant protein production, including bacterial transformation, plasmid isolation, protein expression and purification, gel electrophoresis, and SDS-PAGE.
- Demonstrated and supervised cell lysis, affinity chromatography, and protein quantification, adapting instruction to students' varying skill levels.
- Evaluated laboratory reports and assignments with written feedback on data interpretation, scientific rigor, and technical writing.
- Enforced laboratory safety protocols and aseptic technique in every session.

BIOE 4140: Bioinstrumentation Laboratory

- Led laboratory sessions for 20 to 30 students per semester in the design, construction, and testing of ECG circuits.
- Provided troubleshooting support for circuit design and guided signal interpretation using oscilloscopes and data acquisition tools.
- Evaluated laboratory reports and assignments with feedback on technical accuracy and data presentation.
- Held regular office hours to provide individualized academic support, which contributed to my receiving the 2025 Outstanding Teaching Assistant Award.

BIOE 1410: Freshman Design Innovation (Fall 2025 and Spring 2026)

- Supported first-year students in their first engineering experience, guiding project ideation, design, and development and helping teams resolve technical and conceptual challenges.
- Several students from the course are now pursuing patents for projects that originated in it, including one team whose project the University of Toledo Tech Transfer Office has agreed to pursue through a provisional patent.

MENTORSHIP AND STUDENT TRAINING

Graduate Research Mentor *2021 to Present*

Biofluidics Lab, Department of Bioengineering, University of Toledo

- Mentored four undergraduate students in microfluidic device operation, primary cell culture, fluorescence imaging, and experimental design, enabling independent contributions to ongoing projects.
- Trained two graduate students in tri-culture model setup, cell seeding, and experimental workflows.

- Guided approximately five high school students during summer laboratory visits, providing hands-on exposure to microfluidic research and biomedical engineering career pathways.

PROFESSIONAL SERVICE

Peer Review Assistance *2023 to Present*

Advanced Biology (Wiley); Biotechnology and Bioengineering (Wiley)

- Assisted with manuscript peer review.

AWARDS AND HONORS

Outstanding Teaching Assistant Award *2025*

College of Engineering, University of Toledo. Selected through combined student evaluation and faculty nomination.

Advanced Leadership Academy, Nominated Participant *Spring 2023*

ProMedica and the College of Business, University of Toledo. Nominated cohort program in inclusive leadership, community development, and professional growth.

OUTREACH AND VOLUNTEER ACTIVITIES

University of Toledo Engineering Summer Camp, Bioengineering Lab Session Instructor *July 2026*

University of Toledo, College of Engineering

- Co-led the bioengineering laboratory session with Dr. Kelly Marbaugh for high school students from Ohio, Michigan, and Indiana.
- Guided students through alginate synthesis to demonstrate how material design influences drug delivery.
- Featured in Toledo Blade coverage of the camp as a graduate researcher and science communicator.

Engineer For a Day, Middle School Outreach Instructor *March 2026*

University of Toledo

- Led a hands-on alginate activity in which students used syringes to design their own beads, varying size, shape, and color.
- Connected the activity to biomedical engineering by explaining how alginate is used to study controlled drug delivery, including molecular diffusion and release over time.
- Fostered student curiosity and translated biomedical engineering concepts for young, non-specialist audiences.

Girl Day, Event Coordinator *March 2025*

University of Toledo

- Coordinated student group logistics across multiple campus locations for a university-wide STEM outreach event for girls in K-12 education.
- Managed group transitions, scheduling, and activity coordination throughout the event.

Engineer For a Day, STEM Outreach Presenter *February 2023, February 2024*

University of Toledo

- Delivered 20-minute presentations on microphysiological systems research to rotating groups of 20 to 25 high school students and teachers.
- Represented the Biofluidics Lab at this annual event for two consecutive years.

MEDIA COVERAGE

Press: Featured in "Teens Disassemble Engines, Build Career Visions at UT Engineering Camp," The Toledo Blade (Toledo, Ohio), July 21, 2026.

Video: Featured in the University of Toledo promotional commercial (2026).

PROFESSIONAL MEMBERSHIPS

Biomedical Engineering Society (BMES) | American Physiological Society (APS)

TECHNICAL SKILLS

Microfluidics & microphysiological systems. Microphysiological system (MPS) design and operation; tri-culture microfluidic model development; intestinal microphysiological models; blood-brain barrier model; commercial microfluidic device operation; high-throughput microfluidic assay execution; physiological microenvironment modeling; perfusion culture under defined shear stress.

Cell biology & primary cell culture. Mammalian cell culture; primary cell isolation and culture (rat colonic epithelial cells, F11 DRG sensory neurons); HUVEC culture; BMEC culture; co-culture and tri-culture system development; transient transfection (Lipofectamine 3000); plasmid DNA delivery; aseptic technique; long-term culture maintenance.

Assays & functional readouts. Immunofluorescence; confocal z-stack microscopy; fluorescence microscopy; real-time Fluo-4 AM calcium imaging (single-cell ROI analysis); transepithelial permeability assays; ZO-1 tight junction characterization; barrier integrity assessment; competitive ELISA; cytokine profiling; Western blot; MTT and CellTiter-Blue viability assays; ROS detection; IC50 dose-response pharmacological screening; receptor antagonist and kinase inhibitor studies; hypoxia/HIF modeling; spectrophotometric protein quantification.

Molecular biology. PCR and qPCR; RNA extraction; Western blot; recombinant protein expression (IPTG induction); Ni-NTA affinity chromatography; SDS-PAGE; agarose gel electrophoresis; bacterial transformation; plasmid DNA isolation; bioreactor fermentation.

Software & computational tools. COMSOL Multiphysics (FEA); GraphPad Prism; ImageJ/FIJI; MATLAB; LabVIEW; SolidWorks (CAD); 3D printing; Python; Microsoft Office Suite